Department of Computer Science and Engineering.

Academic Year: 2023- 2024(Odd/~~Even~~ Semester)

Degree, Semester & Branch: B.E,III & EEE
Course Code & Title: CS3353 & C programming and Data Structure
Name of the Faculty member (s): A. Sofia Thanga Mary, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit V / Sorting And Searching Techniques**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO13
  - **Activity Chosen:** Flipped classroom
  - **Justification:** The basics of sorting and searching are very important for understanding the steps involved in searching and sorting. They have already studied in python programming in previous semester. Hence to recollect and get a sound knowledge in the topic by viewing the material and have elaborate discussion in class room, this flipped classroom activity is planned.
- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**
  - ➢ The learning materials were posted to the students in canvas two days before the topic discussion.
  - ➢ The students are instructed to study the material and take notes on the topic based on their understanding.
  - ➢ On the day of the activity the students were grouped into a team of 2 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes and it is shown in Fig.1.
  - ➢ Based on the study material they have gone through, the problems were given for each group separately and by discussing with their team members they have solved it.
  - ➢ After the discussion one member from each team had a chance to present and share the way of solving the problems to the entire class as shown in Fig.2 & 3.
  - **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1	PSO2
	3	3	2	2	1	2	2	2

  - **PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PSO1	PSO2
	3	**3**	**2**	**2**	**2**	**2**	**2**
Justification for correlation	Needs to learn the basic knowledge of engineering fundamentals to perform quick sorting.	The students will identify the relevant sorting algorithm for a given application.	Interpret the complex engineering problem with the help of sorting algorithms	The students can identify best sorting algorithm based on their analysis.	The students can do individually perform quicksort	The students can continued upgrading of technical knowledge using the concept of sorting.	The students able to implement different sorting algorithm in ICT applications

• Images / Screenshot of the practice:

- **Reflective Critique:**
- **Feedback of practice from students and other stakeholders**:
    - ✓ Student felt good, since they can study at their own pace/time.
    - ✓ They felt that through such learning, they can explore more.
- **Benefit of the practice:**
    - ✓ More one-to-one time between teacher and student.
    - ✓ More collaboration time for students.
    - ✓ Students learn at their own pace.
    - ✓ Practical things – like missing class due to illness – become less problematic.
    - ✓ It encourages students to come to class prepared.
- **Challenges faced in implementation:**
    - ✓ Relies on student preparation – few students did not come prepared.
    - ✓ The depth of the subject can be dictated by the student themselves or the group the student is working with.
    - ✓ The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- https://www.teachthought.com/learning/the-definition-of-theflippedclassroom/#:~:text=A%0 flipped%20classroom % 20is%20a,the% 20students%20in dependently%20at%20home.
- https://en.wikipedia.org/wiki/Flipped_classroom
- youtube.com/watch?v=BCIxikOq73Q
- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-whatwhy-and-how-to-implement-a-flipped-classroom-model

Signature of Faculty Member **HOD**